

Integral Inclusion

$$y(t) \in \int_0^T k(t,s)A(s,y(s))ds$$

a.e. $t \in [0, T]$

*Fixed
point*



Multivalued Operator

$$A : C \rightarrow \mathcal{K}(C)$$

(Upper semicontinuous)



Solution Space

$$y \in L^p[0, T]$$

with $\|y\|_p < \alpha$

Key Properties:

- Upper semicontinuous mapping
 - Compact-valued
- Fredholm integral inclusion